

AI for Mathematics

History, Agents, and Formal Methods

Seminar 1

An introduction to the discussion series

Liu Peixuan

SUSTech Math Department

Research snapshot: 9 September 2026

Three questions for the seminar

Part	Question
1. History and the research frontier	What changed between symbolic provers and recent research results?
2. Agents	How does a model turn a suggestion into a sustained workflow?
3. Formal methods	How does type theory support proof checking, and what Lean topics come next?

Discovery, proof construction, and verification are distinct tasks.

History and the research frontier

01

Mechanized reasoning, learned search, and new mathematics

Recall: Hilbert's formalist program

- ▶ **Mathematics as formal systems:** symbols, axioms, inference rules.
- ▶ **Metamathematics:** study proofs as finite mathematical objects.
- ▶ **Original ambition:** finitistic consistency proofs.
- ▶ **Gödel:** limits to that ambition; mechanized proof checking remains useful.

Early progress: from rules to checked proofs

Period	Milestone	What changed
1950s–1970s	Logic Theorist; LCF	Automated deduction; a small trusted core
1976	Appel–Haken: four colors	Computer-assisted case analysis
2005	Gonthier: four colors in Coq	Formal verification of the full argument
2014	Flyspeck: Kepler conjecture	Large-scale formal proof engineering

Sources: [\[4C\]](#) [\[4CP\]](#)

AI for mathematics: beyond generative LLMs

System	Learned component	Role / environment
DeepMath (2016)	CNN / RNN	Premise selection for Mizar problems
TacticToe (2021)	Learned tactic guidance	Monte Carlo tree search in HOL4
Lean (2023)	Random forests	Premise selection
LeanHammer (2025/26)	Encoder-based retrieval	Premises + symbolic provers + Lean reconstruction

Sources: [\[DM\]](#) [\[TT\]](#) [\[RF\]](#) [\[LH\]](#)

AI for mathematics: discovery and hybrid search

Approach	Example	Mathematical output
Learn patterns	Davies et al. (2021)	Relations between invariants; human conjectures and proofs
Neural + symbolic	AlphaGeometry (2024)	Auxiliary constructions; symbolic geometry proofs
Program search	FunSearch (2023)	Cap-set constructions; executable evaluation
Evolution + LLM	AlphaEvolve (2025)	Algorithms and finite constructions

Sources: [INT] [AG] [FUN] [AE]

2025: the Hilbert workflow

A specialized orchestration of complementary models

Stage	Component
Retrieve	Relevant Mathlib theorems
Reason	General LLM: informal argument
Decompose	Lean sketch with smaller subgoals
Prove / repair	Specialized prover + Lean feedback; recurse on failures

Sources: [\[HIL\]](#)

Remark: re-test the old workflow

- ▶ **Working view:** many 2025 pipelines may now lose to a general agent.
- ▶ **A major driver:** stronger base models, not only elaborate orchestration.
- ▶ **Evidence:** unit-distance discovery with a general reasoning model; zeta research through Claude Code.
- ▶ **Fair comparison:** same model, problem set, tools, and compute budget.

Sources: [\[UD\]](#) [\[ZETA\]](#) [\[HIL\]](#)

AI4Math results: May–August 2026

Release	Reported advance	Route / evidence
20 May	Unit distance: disproof of $u(n) = n^{1+o(1)}$	Natural-language discovery; external expert checking
May	AlphaProof Nexus: Erdős / OEIS problems	Lean-target search; exact variants matter
July	Jacobian conjecture: counterexample in $\mathbb{C}^3$	Explicit construction; algebraic checks
1 Aug	Ten advances: packing, non-sofic groups, Ramsey, etc.	Arguments first; Lean certificates afterward
10 Aug	Zeta zeros: $\kappa \geq 0.672$	Claude Code; expert examination + Lean

Sources: [UD] [NEX] [ZETA] [TEN]

AI4Math results: the latest releases

Release	Reported advance	Evidence / scope
August	Complex structure on S^6	Manuscript; Engel exposition; Lean release
Late Aug	Critical bond percolation: $\theta(p_c) = 0$ on $\mathbb{Z}^d$, $d \geq 2$	Lean release and audit; independent review pending in snapshot
4 Sep	Fermat's Last Theorem	Formalization milestone; 11 days, 13M Lean lines
7 Sep	Alpöge–Buckmaster: smooth-forced fluid blowup	IPM, Boussinesq, Euler; papers + Lean
8 Sep	OpenAI: smooth-forced Navier–Stokes blowup	Paper + Lean; claimed Clay C/D resolution

Sources: [\[S6\]](#) [\[ENG\]](#) [\[PERC\]](#) [\[FLT\]](#) [\[BA\]](#) [\[NS\]](#)

Remark: formalization and doing mathematics

- ▶ **Recent pattern:** natural-language discovery, then formalization.
- ▶ **Lean's role:** stronger confidence in the encoded claim.
- ▶ **Beautiful mathematics too:** type theory, HoTT, reusable abstractions.
- ▶ **My concern:** over-focusing on formalization may miss the mathematical question.

A caution from symbolic AI: representation alone does not choose a fruitful idea.

Sources: [TEN] [ZETA] [WHY]

Levent Alpöge: a mathematician using agents

When	Documented trajectory
Before AI	Number theory; Princeton PhD; Harvard Society of Fellows
2023 onward	GPT-4 motivates a return to CS; joins Anthropic
2025–2026	Extended LLM-assisted collaboration with Buckmaster
June–August 2026	Public Claude Code experiments; Jacobian / S^6 ; zeta proof examination

Sources: [LEV] [LEVCC] [BA] [ZETA]

The NS episode: timeline

Date	Publicly reported event
15 / 22 Aug	Buckmaster: forced Euler / Boussinesq proof, then Lean check
1 Sep	OpenAI: evaluation effort begins after rumors
3 / 6 Sep	Private outreach, then publication / authorship discussions
7 Sep	Alpöge–Buckmaster papers and statement become public
8 Sep	OpenAI releases NS paper + Lean; Bubeck responds

Sources: [\[BA\]](#) [\[NS\]](#) [\[SB\]](#)

The NS episode: what is disputed?

Issue	Buckmaster's account	Response / open point
Data	Drafts entered into Codex	OpenAI denies targeted access; training influence not ruled out
Authorship	Proposed NS writeup excluded Alpöge	Bubeck: own-work credit vs authoring OpenAI's proof
Pressure	Career-related remarks	Bubeck apologizes for wording; disputes intent

Unresolved: provenance and fair credit. A valid proof cannot settle either.

Sources: [\[BA\]](#) [\[NS\]](#) [\[SB\]](#)

Resources: who owns the means of research?

~10,000

Concurrent agents

OpenAI's reported NS group

- ▶ **Models:** frontier internal versions; access beyond the public tier.
- ▶ **Compute:** large parallel searches; repeated attempts.
- ▶ **Data:** platform control over processing and permitted model improvement.
- ▶ **Structural tension:** tool provider and research competitor.

Sources: [NS]

Objectives: what counts as progress?

Company incentives

Benchmark performance; capability demos

Product demand; capital / IPO narrative?

Fast, recognizable breakthroughs

Mathematical values

Understanding; concepts; useful methods

Credit; students; a research community

The glory of the human mind?

Tao: solved problems as an imperfect proxy for progress.

Hugo: exploration and becoming a mathematician.

Sources: [\[TAO26\]](#) [\[HUGO\]](#) [\[NS\]](#)

Consequences: from service providers to publishers

- ▶ **Already happening:** companies publish mathematical claims directly.
- ▶ **Possible next step:** routine publication without academic mediation.
- ▶ **Pressure points:** priority, private ideas, review capacity, young researchers.
- ▶ **Precedent:** Copilot training and the 2022 Give Up GitHub campaign.

Who sets the agenda, receives credit, and gets to participate?

Sources: [TEN] [NS] [TAO26] [HUGO] [GH]

How general-purpose agents work

02

Reasoning, tools, and an iterative control loop

Agent = LLM + harness

$$\text{agent} = \text{LLM} + \text{harness}$$

- ▶ **LLM:** reasoning, plans, code, tool-call proposals.
- ▶ **Harness:** tools, context / memory, permissions, execution loop.
- ▶ **Feedback:** execute, observe, revise, continue or stop.

Chain of thought: model-generated reasoning

1. The model receives the current conversation.
2. It generates output one token at a time.
3. A reasoning model can work through intermediate steps before an answer or tool call.

All inside model generation.

No external program runs at this stage.

A displayed “thinking” summary may differ from the underlying reasoning.

Sources: [\[CODEX\]](#) [\[COT\]](#)

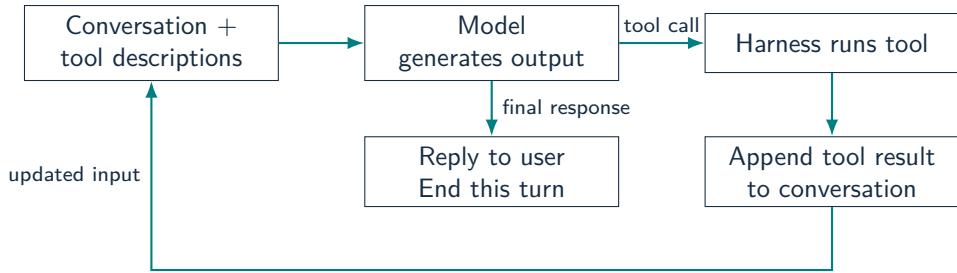
Tool calling: output, execute, return

1. **Model outputs:** tool name + arguments.
2. **Harness executes:** the requested operation.
3. **Harness returns:** output or error, added to the conversation.
4. **Model continues:** reads that result in its next call.

A tool call is a request. The harness makes it happen.

Sources: [\[CODEX\]](#)

The agent loop



One user request can involve many model calls and tool runs.

Sources: [\[CODEX\]](#) (adapted)

Example: a SageMath calculation

User: Factor $x^4 - 1$ over $\mathbb{F}_5$.

Model: writes `factor.sage`; requests `shell("sage factor.sage")`.

```
R = PolynomialRing(GF(5), 'x')
x = R.gen()
print((x^4 - 1).factor())
```

Harness: runs SageMath; returns the output to the model.

```
(x + 1) * (x + 2) * (x + 3) * (x + 4)
```

Model: reports four linear factors in $\mathbb{F}_5[x]$.

Sources: [\[SAGE\]](#)

Suggested AI tools

Provider	Model family	Agent interface
OpenAI	GPT	Codex
Anthropic	Claude	Claude Code
Z.ai	GLM	ZCode
Moonshot AI	Kimi	Kimi Code
Alibaba	Qwen	Qwen Code

Sources: [\[CODEX\]](#) [\[CC\]](#) [\[GLM\]](#) [\[KIMI\]](#) [\[QWEN\]](#)

03

Curry–Howard, dependent types, and our next topics

Curry–Howard Correspondence: propositions as types

$\Gamma \vdash p : P$ means that p is a proof of P in context Γ .

Logical form	Type-theoretic reading
$P \rightarrow Q$	A function transforming a proof of P into a proof of Q
$\forall x : A, P(x)$	A dependent function assigning a proof to each x
$P \wedge Q$	A constructor containing proofs of both propositions
$\exists x : A, P(x)$	A witness and a proof, packaged in an existential proposition

In Lean, `Prop` has proof irrelevance; `Exists` in `Prop` is distinct from a data-valued Σ type.

Sources: [\[DTT\]](#) [\[PROP\]](#)

Curry–Howard: a proof is a typed term

Logical claim: from $P \wedge Q$, derive $Q \wedge P$.

$$\frac{h : P \wedge Q}{\langle h.2, h.1 \rangle : Q \wedge P} \rightsquigarrow \lambda h. \langle h.2, h.1 \rangle : (P \wedge Q) \rightarrow (Q \wedge P).$$

```
example (P Q : Prop) : P /\ Q -> Q /\ P :=  
  fun h => And.intro h.right h.left
```

- ▶ Assuming a proposition corresponds to introducing a typed variable.
- ▶ Using a proof corresponds to applying or inspecting a term.
- ▶ Checking the proof corresponds to checking the term's type.

Sources: [\[PROP\]](#)

Recall: dependency, equality, and induction

- ▶ **Dependent functions:** $f : \prod_{x:A} B(x)$; the output type depends on the input.
- ▶ **Definitional equality:** computation identifies $(\lambda x. t) a$ with $t[a/x]$.
- ▶ **Propositional equality:** $h : a = b$ is a proof object used for rewriting and transport.
- ▶ **Inductive types:** constructors determine elimination principles; induction builds proofs over structured objects.

A tactic is a procedure for building a proof term; the kernel checks the resulting term.

Sources: [\[DTT\]](#) [\[PROP\]](#)

Why formalization? Correctness

Formalize a theorem, or a proof?

Question	What must be checked?
Is the theorem true?	Formal statement matches the intended claim
Was my proof correct?	Formal argument follows the original reasoning
Does Lean accept it?	A proof term, with its actual axioms and dependencies

Same theorem, different proof: $\sqrt{2} \notin \mathbb{Q}$ via parity or FLT(3).

Sources: [\[WHY\]](#)

Why formalization? Generalization for (almost) free

- ▶ Change a constant, hypothesis, or ambient structure.
- ▶ Re-run the proof; inspect exactly where it breaks.
- ▶ Reuse the surviving lemmas and abstractions.

Example: Polynomial Freiman–Ruzsa

$$2K^{12} \longrightarrow 2K^{11}.$$

Improvement by Liao; existing formalization helps carry it through.

Sources: [\[WHY\]](#)

Why formalization? Understanding thoroughly

- ▶ Expose hidden assumptions and actual dependencies.
- ▶ Find the right definitions; extract reusable lemmas.
- ▶ Replace a difficult detour with a simpler argument.

Example: sphere packing formalization

A direct argument replaces a technically expensive dependency.

Read the proof code; understand what was formalized.

Sources: [\[WHY\]](#)

A small example in Lean

Example: every square of an odd natural number is odd.

```
import Mathlib

theorem odd_square_explicit (k : Nat) :
  (2*k + 1)^2 = 2*(2*k^2 + 2*k) + 1 := by
  ring
```

The identity supplies the witness $2k^2 + 2k$ for oddness.

Scope: the identity makes the witness explicit.

A small certificate is useful when its exact scope is explicit.

Upcoming Lean topics

For subsequent sessions

1. `type_conversion`
2. Sets & Functions
3. Elementary Algebra
4. Group Theory
5. Ring Theory
6. Field Theory
7. Matrix Theory

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